

Assignment No. <4.1>

<Switch case >

Course Code: CPE007

Program: Computer Engineering

Course Title: Programming logic and design

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Name(s): Hanz Rhayven M, Pabalan

Instructor: Engr. Jimlord Quejado

6. Output

Output and Code:

main.cpp

Share

Run

```
1  #include <iostream>
2  using namespace std;
3- int main(){
4      int physics, biology, math;
5      float average;
6      char grade;
7
8      cout<<"Enter marks for physics: ";
9      cin>>physics;
10     cout<<"Enter marks for biology: ";
11     cin>>biology;
12     cout<<"Enter marks for math: ";
13     cin>>math;
14
15     average = (physics + biology + math) / 3.0;
16-    switch ((int)average / 10) {
17        case 10:
18        case 9:  grade = 'A';
19        break;
20        case 8:  grade = 'B';
21        break;
22        case 7:  grade = 'C';
23        break;
24        case 6:  grade = 'D';
25        break;
26        case 5:
```

Output

Enter marks for physics: 89
Enter marks for biology: 80
Enter marks for math: 87

Physics : 89
Biology : 80
Math : 87

Average is : 85.3333
Grade Level: B

=== Code Execution Successful ===

main.cpp

Share

Run

```
21     break;
22     case 7:  grade = 'C';
23     break;
24     case 6:  grade = 'D';
25     break;
26     case 5:
27     case 4:  grade = 'E';
28     break;
29     default: grade = 'F';
30     break;
31 }
32
33 cout << "\nPhysics : " << physics << endl;
34 cout << "Biology : " << biology << endl;
35 cout << "Math : " << math << endl;
36 cout << "_____ " << endl;
37 cout << "Average is : " << average << endl;
38 cout << "Grade Level: " << grade << endl;
39     return 0;
40 }
41
```

Output

Enter marks for physics: 89
Enter marks for biology: 80
Enter marks for math: 87

Physics : 89
Biology : 80
Math : 87

Average is : 85.3333
Grade Level: B

=== Code Execution Successful ===

Pseudocode:

START

DECLARE physics, biology, math, english, chemistry AS INTEGER
DECLARE total, average AS FLOAT
DECLARE grade AS CHARACTER

DISPLAY "Enter marks for Physics:"
INPUT physics
DISPLAY "Enter marks for Biology:"
INPUT biology
DISPLAY "Enter marks for Math:"
INPUT math
DISPLAY "Enter marks for English:"
INPUT english
DISPLAY "Enter marks for Chemistry:"
INPUT chemistry

total $\leftarrow$ physics + biology + math + english + chemistry
average $\leftarrow$ total / 5

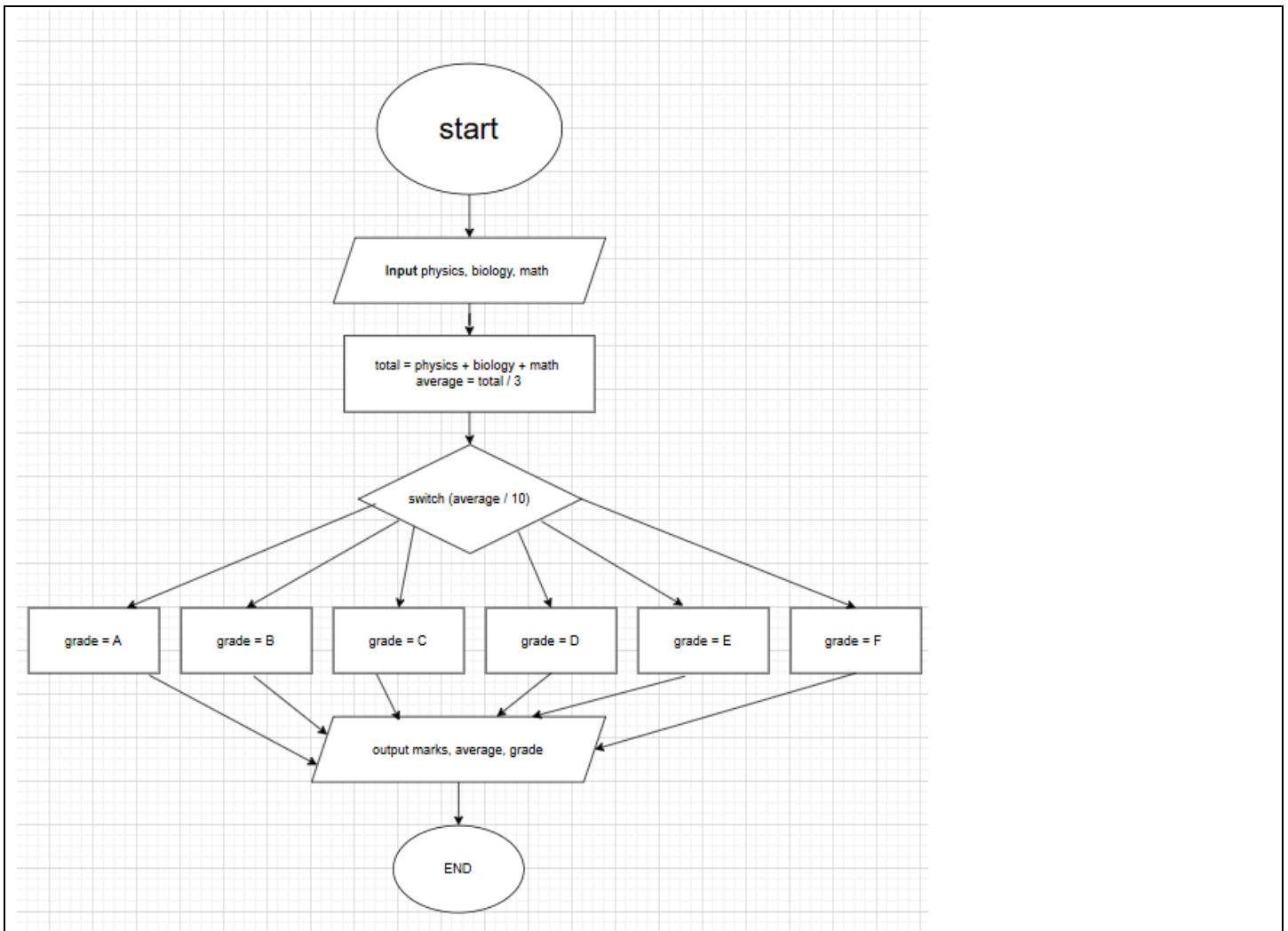
SWITCH (average DIV 10)

CASE 10, 9:
 grade $\leftarrow$ 'A'
CASE 8:
 grade $\leftarrow$ 'B'
CASE 7:
 grade $\leftarrow$ 'C'
CASE 6:
 grade $\leftarrow$ 'D'
CASE 5, 4:
 grade $\leftarrow$ 'E'
DEFAULT:
 grade $\leftarrow$ 'F'
END SWITCH

DISPLAY "Physics : ", physics
DISPLAY "Biology : ", biology
DISPLAY "Math : ", math
DISPLAY "English : ", english
DISPLAY "Chemistry : ", chemistry
DISPLAY "_____"
DISPLAY "Average is : ", average
DISPLAY "Grade Level: ", grade

END

FLOWCHART:



7. Supplementary Activity

8. Conclusion

This task helped a lot to practice my skills, practice my switch case and understand the connection of programming and visual representation, especially now that I'm struggling on my flowcharts.

9. Assessment Rubric